

Self-regenerating zwitterionic hyperbranched polymer with tunable degradation for anti-biofouling coatings

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ABSTRACT

Regeneration of zwitterionic surfaces is a promising strategy to combat biofouling. In this study, we have prepared degradable hyperbranched anti-biofouling polymers of methyl methacrylate, tertiary carboxybetaine ester (TCB), and divinyl monomers via reversible addition-fragmentation chain transfer (RAFT) polymerization. The degradation rate of such polymers is a function of the content or type of divinyl monomers. As the content of methacrylic anhydride (MAAH) increases, the degradation rate of the polymer becomes faster; whereas, a change of the divinyl monomers causes the degradation rate to follow a descending order: *N,N'*-adipic bis(diacetone acrylamide hydrazone) (DAA₂H) > MAAH > ethylene glycol dimethacrylate (EGDMA). Antibacterial and anti-diatom assays show that the coatings inhibit the adhesion of the marine bacteria *Pseudomonas* sp. and the diatom *Navicula incerta*. The polymeric coating is expected to be applied in marine anti-biofouling.

1. Introduction

Marine biofouling is caused by the accumulation of microbial biofilm on artificial surfaces such as ship hulls followed by progressively larger marine organisms, which may cause serious economic and ecological problems [1–3]. To combat biofouling, many efforts have been made to develop coatings that kill or inhibit its adhesion. However, most coatings that kill biofouling contain antifoulants, and the released antifoulant may result in secondary pollution of the marine environment. Thus, it is significant to develop environmentally friendly and non-toxic biofouling control strategies [4–6].

Several anti-biofouling coatings have been explored, including fouling release coatings [7,8], biomimetic coatings [9,10], and protein-resistant coatings [11–13]. Among them, protein-resistant coatings attract more attention and are expected to exhibit good anti-biofouling performance because they prevent the adsorption of proteins, thereby inherently suppressing the colonization of bacteria and marine microorganisms on their surfaces. The most frequently studied is poly(ethylene glycol) (PEG) or zwitterionic polymers. However, PEG has poor stability because its macromolecular chains are prone to autoxidation and degradation, especially by transition metal ions [14,15]. Zwitterionic polymers are a class of materials with a unique molecular structure. Their polymer chains have the same number of cationic and

anionic groups, which can form a strong and stable hydration layer with superhydrophilicity, preventing the adhesion of proteins on the surface, thus inhibiting the growth of biofouling. However, this also causes coating swelling and poor mechanical properties, which limits their applications. Moreover, zwitterionic polymers gradually lose their anti-biofouling properties because their surfaces may be covered with inorganic substances or dead bacteria in complex marine environments [16–20]. Therefore, designing a self-regenerating or dynamic surface is a promising method that can discard a surface layer with the attached fouling and regenerate a new anti-biofouling surface [21–23].

Degradable polymers, which can be degraded into small molecules to form a self-renewing surface, are candidates for marine anti-biofouling applications [7,24–27]. However, it is difficult to find suitable co-solvents to dissolve highly polar zwitterions and degradable polymers. To overcome this problem, zwitterion precursors have been developed and incorporated into degradable polymer, which are soluble in most organic solvents, and generate zwitterions that inhibit protein adhesion under water attack [28,29]. Particularly, combining zwitterion precursors with degradable hyperbranched polymers (HBPs) can form surface-fragmenting, self-regenerating anti-biofouling surfaces, which could underpin the next generation of anti-biofouling solutions since HBPs have good solubility, low viscosity, and abundant end groups relative to linear analogs. Reversible addition-fragmentation chain

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transfer (RAFT) polymerization is a facile and versatile approach to synthesize HBPs due to its many advantages such as relatively mild reactions, wide range of monomers, and tolerance of various functionalities. Usually, the RAFT polymerization of HBPs consist of vinyl monomers, di- or multivinyl monomers and chain transfer agent [30–32]. It is worth noting that the selection of di- or multivinyl monomers is important as they affect the self-renewing and anti-biofouling ability of coatings [33–38].

Herein, we have synthesized degradable HBPs with controlled structure of methyl methacrylate (MMA), tertiary carboxybetaine ester (TCB), and divinyl monomer-methacrylic anhydride (MAAH) via RAFT polymerization. Among them, the TCB exhibits hydrolysis-induced zwitterionic characteristics and excellent non-fouling properties; whereas, MAAH moieties provide degradation performance. In addition, we examined the effect of divinyl monomers including ethylene glycol dimethacrylate (EGDMA), and *N,N'*-adipic bis(diacetone acrylamide hydrazone) (DAA₂H) on polymer properties. We have studied the wettability, degradability, and anti-biofouling properties of the polymers and correlated these properties with their chemical structure and composition. Our goal is to develop an environmentally friendly anti-biofouling coating.

2. Experimental section

2.1. Materials

Diacetone acrylamide (DAA, Aladdin, 99%) was recrystallized from hexanes. 2,2-Azobisisobutyronitrile (AIBN, 98%, Aladdin) was recrystallized twice from methanol. Methyl methacrylate (MMA, 99%, Aladdin) was purified by distillation before use. 2-Cyano-2-propyl dodecyl trithiocarbonate (CPDT, 95%) was purchased from Zhengzhou JACS Chem. Ethylene glycol dimethacrylate (EGDMA, 98%), methacrylic anhydride (MAAH, 94%), adipic dihydrazide (99%), ethyl acrylate (99%), acryloyl chloride (96%), 2-methylaminoethanol (99%), and triethylamine (TEA, 99%) were purchased from Aladdin. Anhydrous tetrahydrofuran (THF, 99.5%) and anhydrous dimethylformamide (DMF, 99.8%) were purchased from Energy Chemical. Glacial acetic acid (AR) and ether (AR) were purchased from Guangzhou Chemical Reagent. Tertiary carboxybetaine ester (TCB) was synthesized based on a previously reported procedure [28]. *N,N'*-adipic bis(diacetone acrylamide hydrazone) (DAA₂H) was synthesized according to a procedure

described elsewhere [37]. The ¹H NMR spectra of MMA, TCB, three kinds of divinyl monomers and RAFT agent can be found in Fig. S1. Artificial seawater was prepared following the ASTM D1141-98 (2013) standard.

2.2. Synthesis of hyperbranched polymers

The hyperbranched polymers were prepared by RAFT polymerization of MMA, TCB, and divinyl monomers (Scheme 1). Typically, a mixture of MMA (0.84 g, 8.40 mmol), TCB (0.86 g, 3.75 mmol), MAAH (0.058 g, 0.38 mmol), CPDT (0.108 g, 0.31 mmol), and AIBN (0.010 g, 0.06 mmol) was dissolved in anhydrous DMF (4.00 g) and added into a Schlenk flask with a magnetic stirring bar. Then, the mixture was degassed using three freeze-evacuate-thaw cycles and was backfilled with nitrogen. The polymerization was conducted at 70 °C for 24 h. Finally, the polymer was precipitated in ether and dried overnight under vacuum at 60 °C. For all polymerizations, the feed molar ratio of [MAAH] to all monomers was varied from 0 to 6%, and the molar ratio of [CPDT]:[AIBN] was maintained at 2.5:0.5. For convenience, the polymers were named as *h*-Y_x, where Y represents EGDMA, MAAH or DAA₂H, and x is the feed molar ratio of Y. The characterization data are shown in Table 1.

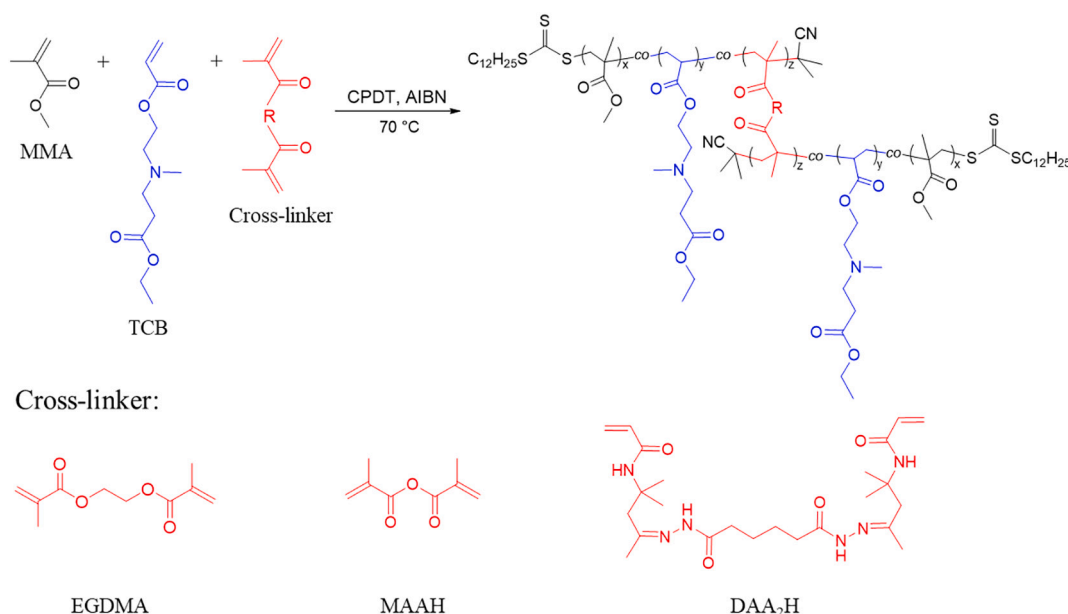
3. Characterization

3.1. Proton nuclear magnetic resonance spectroscopy (¹H NMR)

All ¹H NMR spectra were obtained using a Bruker AV600 NMR spectrometer. The samples were dissolved in CDCl₃ except the DAA₂H sample dissolved in DMSO-*d*₆. All resonance bands use tetramethylsilane (TMS) as an internal standard.

3.2. Gel permeation chromatography (GPC)

Using two identical PLgel MIXED-C columns with a flow rate of 1.0 mL/min, the number-average molar mass (*M_n*) and the polydispersity (*Đ_M*) of the polymers were determined in THF at 35 °C. A series of polystyrene standards with narrow polydispersity were used for calibration.



Scheme 1. Synthesis of hyperbranched polymers by RAFT polymerization.

Table 1

Characterization data of hyperbranched polymers.

Sample	Feed molar ratio ^a	Divinyl monomer	Polymer composition ^b	M_n^c (g/mol)	M_w^c (g/mol)	$M_{n,theo}^d$ (g/mol)	D_M^c	DB ^e (mol%)	T_g^f (°C)
h-MAAH0	70/30/0	MAAH	71.5/28.5	7800	11,000	5900	1.41	0	5.1
h-MAAH3	67/30/3	MAAH	67.6/29.0/3.4	8700	14,300	5700	1.64	3.4	13.8
h-MAAH6	64/30/6	MAAH	64.7/28.9/6.4	9000	15,500	5500	1.72	6.4	27.1
h-EGDMA3	67/30/3	EGDMA	69.4/27.4/3.2	13,000	30,000	5800	2.30	3.2	16.3
h-DAA ₂ H3	67/30/3	DAA ₂ H	68.5/28.4/3.1	6000	9200	5700	1.53	3.1	22.8

^a Feed molar ratio of MMA/TCB/Divinyl monomer. [CPDT]/[AIBN] = 2.5/0.5 for all polymerizations.^b Polymer composition was determined by ¹H NMR. The details of calculation and the relative reactivity ratio between MMA and TCB can be found in Supporting Information.^c Obtained from GPC.^d Theoretical number average molecular weight of linear segments between branching points, $M_{n,theo} = ([MMA]/[CPDT]) \times \text{conversion} \times M_{W, MMA} + ([TCB]/[CPDT]) \times \text{conversion} \times M_{W, TCB} + M_{W, CPDT}$.^e Degree of branching (DB) was calculated from the equation: $DB = D/(D + L)$, where D is the number of branching units and L is the number of linear units (MMA and TCB) in hyperbranched polymers determined by ¹H NMR.^f Determined by DSC.

3.3. Differential scanning calorimetry (DSC)

DSC measurement was performed using Netzsch DSC 204F1 to obtain the glass transition temperature (T_g) at a nitrogen flow rate of 50 mL/min. The sample was quickly heated to 100 °C and held at the temperature for 5 min to eliminate the thermal history. Furthermore, the sample was cooled to −40 °C at the rate of 20 °C/min, and then heated to 100 °C at the same rate. The T_g was determined from the midpoint of the endothermic shift in the second measurement curve. All polymers are amorphous (Fig. S2). The data are summarized in Table 1.

3.4. Adhesion test

The adhesion strength of the coatings on epoxy panel substrate was measured by a pull-off adhesion tester (PosiTest AT-A Automatic). The adhesion strength data were obtained by detaching an aluminum dolly (20 mm in diameter) at a speed of 0.2 MPa/s at 25 °C. Five parallel tests were conducted to obtain an average value.

3.5. Water contact angle (WCA) measurements

The WCA measurements of the polymer films were performed at 25 °C using a Theta Auto 113 (KSV NIMA, biolin Scientific) by dropping 3 μL of deionized water on a polymer surface using a sessile method. Before the measurements, samples were taken out of the artificial seawater at certain intervals, rinsed with deionized water, and dried with nitrogen. Five areas of each sample were measured to obtain the average value.

3.6. Attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy

The ATR-FTIR spectra of the polymer films with different immersion time in artificial seawater at 25 °C were recorded by a Bruker Vertex70 infrared spectrometer at a spectral resolution of 4 cm^{−1} with 32 scans.

3.7. Hydrolytic degradation and water adsorption

The polymer film was prepared by casting on an epoxy resin panel (20 × 20 mm²).

Before immersion in artificial seawater, the weight of each panel was recorded as W_p , and each dried sample together with its panel was measured as W_0 . The sample was immersed in artificial seawater at 25 °C. At certain intervals, the sample was removed from artificial seawater and carefully washed with deionized water. After drying with nitrogen, the weight was recorded as $W_{t,wet}$, then the sample was dried to constant weight in a vacuum freeze dryer to remove the adsorption water, and recorded as $W_{t,dry}$. The mass loss was calculated from ($W_0 -$

$W_{t,dry}$)/($W_0 - W_p$), while the water adsorption was calculated using ($W_{t,wet} - W_{t,dry}$)/($W_{t,dry} - W_p$). Three parallel sample tests were performed on each polymer to obtain an average value.

3.8. Surface morphology measurements

The surface morphology was observed using a 3D optical profiler (RTEC Up Dual Model). The instrument uses white light as the light source, for scanning, to establish the 3D profile of the surface. The sample was prepared by the solution casting method. After a certain immersion time in artificial seawater, the sample was taken out and rinsed gently with deionized water, and then dried at room temperature. Five areas for each sample were scanned to obtain the average surface roughness.

3.9. Antibacterial assays

The antibacterial performance of films was evaluated using marine bacteria *Pseudomonas* sp. NOV77. The film was prepared on a silicon slide (10 × 10 mm²) using spin casting. The bacterial concentration was adjusted to 10⁸ cells/mL with artificial seawater, and each sample was immersed in 1 mL of bacterial suspension for 4 h. Then, the sample was washed slowly by artificial seawater and stained with a LIVE/DEAD BacLight Bacterial Viability Kit. The adhered bacterial cells were observed under a fluorescence microscope (Scope A1, Zeiss). ImageJ software was used to determine the relative bacterial adsorption capacity (RBA), and the average number of bacteria on the silicon wafer was used as a control.

3.10. Anti-diatom assays

The marine diatom *Navicula incerta* was obtained from Xiamen University. The cultivation of diatoms follows the procedures elsewhere [8]. Each sample was immersed in a 24-well plate with 1 mL of diatoms suspension for 24 h. Then, removed the sample and gently rinsed it with artificial seawater to remove unattached diatoms. The attached diatoms were observed with a microscope (Scope A1, Zeiss) and counted through 10 random fields of each sample. Each sample has three parallel samples, and the silicon wafer was the control.

4. Results and discussion

RAFT polymerization was chosen for the synthesis of HBPs (Scheme 1). We initially prepared HBPs with MAAH as branching point. As shown in Fig. S3(a), the typical peak at 3.68–3.50 (c) ppm is attributed to the protons in the side methyl groups from MMA. Peaks at 4.30–3.90 (a, b), 2.80–2.20 (e, f, g, h), and 1.28–1.22 (k) ppm can be all assigned to TCB. The peaks of C=C at 6.0–5.0 ppm disappear. The peak at 0.87 ppm is

attributed to the methyl groups in MAAH, and it increases with the MAAH content. Moreover, the absorption peak at 1805 cm^{-1} in Fig. S4 (a) attributed to the asymmetric stretching vibration of $\text{C}=\text{O}$ from MAAH can be observed. All these results clearly indicate that MAAH-based HBPs have been successfully synthesized. To study the effect of different branching points on polymer properties, the unclavable divinyl monomer EGDMA and another degradable divinyl monomer DAA₂H were used to prepare the control samples. For *h*-EGDMA3, peak at 4.20–4.00 ppm also contains the proton d in EGDMA, which can be confirmed by the area ratio of $A_{(4.20-4.00)}/A_{(2.32-2.22)}$. For *h*-DAA₂H3, the peak at 1.46 ppm can be assigned to the proton k in DAA₂H (Fig. S3(b)). The absorption peak at 1660 cm^{-1} ($\text{C}=\text{N}$) and 1530 cm^{-1} ($\text{C}-\text{N}$) can be observed (Fig. S4(b)). In other words, *h*-EGDMA3 and *h*-DAA₂H3 have been successfully synthesized. Note that though the samples were prepared via RAFT polymerization, their dispersity are relatively broader than that of linear polymers. This is understandable that HBPs have hyperbranching structures, which is totally different from the linear ones. Besides, the hydrodynamic volume of HBPs varies from the content and types of divinyl monomer, which can also affect the molecular weights and dispersity obtained from the GPC. Actually, many other HBPs prepared via RAFT polymerization have been reported to have a broader dispersity even higher than 3.0 [33,40,41].

We observed that when [MAAH] increased from 0 to 6.0, the T_g of *h*-MAAHx gradually increased from 5.1 to 27.1°C , indicating that the T_g of the polymers was positively correlated with the content of divinyl monomers. The increase in the T_g is expected, as the incorporation of the MAAH moieties into the polymer will reduce the mobility of the polymer chains through hydrogen bond interactions [39]. In this study, the divinyl monomers molar ratio of HBPs varies from 0 to ~ 6.0 . Actually, we can easily prepare HBPs with $\sim 10\text{ mol}\%$ of divinyl monomers. It still does not gel in the synthetic process. In addition, the T_g of the polymer is also affected by the type of divinyl monomers. For example, DAA₂H has more complex structure and relatively larger steric hindrance than EGDMA. Thus, the polymer chains flexibility of *h*-DAA₂H3 is lower than that of *h*-EGDMA3, resulting in a higher T_g .

Considering that the marine environment is complicated and the service condition is harsh, we first measured the adhesion strength of HBPs on epoxy substrate after immersion in artificial seawater for different time (Fig. 1). Before immersion, the adhesion strength of each coating is higher than 1.0 MPa because of the high content of acrylate segments in polymer chains, which provides high polar interaction and hydrophobic interaction with the substrate. For *h*-MAAHx, the adhesion strength decreases as MAAH content increases. As indicated by the T_g in Table 1, the incorporation of MAAH lowers the flexibility of polymer chains, thus reducing the contact of esters in polymer chains to the substrate forming more hydrogen bonds between them. Note that *h*-

DAA₂H3 has a highest adhesion strength, this is because it can form more hydrogen bonds between the polymers and the substrate due to the NH groups in DAA₂H. After immersion in artificial seawater, the adhesion strengths have no obvious difference from the initial values. Actually, to accelerate the process of hydrolysis, 0.1 M NaOH was also used (Fig. S5). Similar behavior can be observed. This is because the hydrolysis of TCB and the degradation of branching points only occurred on the surface of coating, and the matrix was not affected. Anyhow, each coating has an adhesion strength of above 1.0 MPa, which can meet the requirement for the marine applications.

Fig. 2 shows the time dependence of water contact angle (WCA) of *h*-Yx surfaces after immersion in artificial seawater. For *h*-MAAHx samples (Fig. 2a), the initial WCA of all samples decreases with the increase in MAAH content. Since all samples have similar TCB contents, the increase in MAAH moieties in the polymer provides more anhydride groups on the coating surface, thereby increasing its hydrophilicity [42,43]. The WCA of *h*-MAAH0 is $\sim 83^\circ$, and the WCA of *h*-MAAH3 is $\sim 79^\circ$. The introduction of more MAAH leads to lower WCA, the WCA of *h*-MAAH6 reached $\sim 75^\circ$. The WCA of all samples gradually decreases as time increases. For *h*-MAAH0, the WCA decreases from 83 to 73° in 7 days and gradually levels off with the immersion time, indicating that the hydrophilic zwitterionic groups are generated on the coating surface. For *h*-MAAH3, the WCA gradually decreases from 79 to 67° in 28 days. The WCA of *h*-MAAH6 sharply decreases from 75 to 63° in 14 days and reaches a constant WCA of $\sim 60^\circ$. Although all the samples have similar TCB contents, the WCA of *h*-MAAH3 and *h*-MAAH6 decreases more rapidly than that of *h*-MAAH0. This decrease is because the incorporation of MAAH in the polymer makes it more hydrophilic, and the anhydride groups in the MAAH moieties can be easily attacked by artificial seawater resulting in a more hydrophilic surface. As shown in Fig. 2b, the initial WCA of *h*-EGDMA3, *h*-MAAH3, and *h*-DAA₂H3 are ~ 78 , 79 , and 81° , respectively, which indicates that the type of divinyl monomers has slight effect on the initial WCA of the polymer. This is probably because the three kinds of divinyl monomers we used in this study have comparable structure and hydrophilicity. After immersion in artificial seawater for 7 days, the WCA of *h*-EGDMA3 slightly decreases and reaches a constant of $\sim 72^\circ$, since EGDMA does not degrade and only the hydrolysis of TCB provides hydrophilicity. For *h*-MAAH3, the WCA is always lower than that of *h*-EGDMA3 after immersion in artificial seawater, because the hydrolysis of one anhydride group in MAAH moieties can produce two carboxyl groups, resulting in a more hydrophilic surface. Similarly, the WCA of *h*-DAA₂H3 decreases to $\sim 71^\circ$ because of the hydrolysis in TCB and the generation of hydrophilic ketone carbonyl group on the surface. Note that the WCA of *h*-DAA₂H3 is much greater than that of *h*-MAAH3, this is probably because the hydrazone bond in DAA₂H is hydrolyzed to generate two ketone carbonyl groups, but the hydrophilicity of ketone carbonyl group is less than carboxyl group [44]. Anyhow, the fact clearly indicates the hydrolysis of TCB and the degradation of branching points.

We examined the hydrolysis of TCB groups in the polymers in artificial seawater by ATR-FTIR. Fig. 3a, b, and c shows the ATR-FTIR spectra of *h*-EGDMA3, *h*-MAAH3, and *h*-DAA₂H3 hydrolyzed in artificial seawater from 0 to 28 days. As the immersion time in artificial seawater increases, the new peaks were observed to become more obvious in all three samples. For all samples, the new peak was observed at 1560 cm^{-1} , indicating that TCB is converted into carboxybetaine groups. In addition, the broad peak at 3425 cm^{-1} is attributed to the water molecules adsorbed on hydrophilic zwitterionic groups, while the peak at 1640 cm^{-1} can be assigned to the carboxyls converted from the esters. These results confirm the generation of zwitterionic surfaces. We also characterized the ATR-FTIR spectra of *h*-MAAH0 (Fig. S6(a)) and *h*-MAAH6 (Fig. S6(b)) immersed in artificial seawater from 0 to 28 days. Compared with *h*-MAAH0, the additional peak at 1805 cm^{-1} in *h*-MAAH3 and *h*-MAAH6 were attributed to the carbonyl of the anhydride group, further indicating that the MAAH moieties were successfully incorporated into the polymer [45,46]. Although the samples have

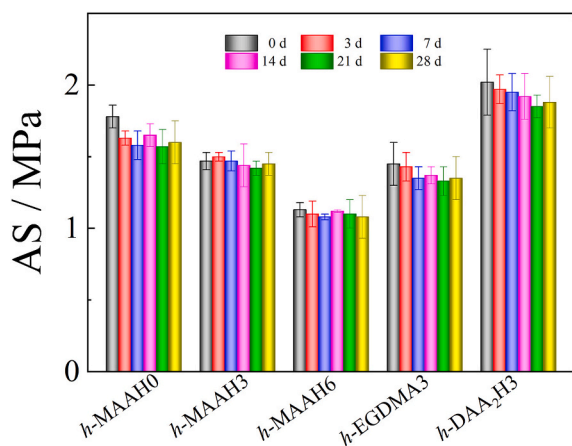


Fig. 1. The adhesion strength (AS) of HBPs on epoxy substrate after immersion in artificial seawater for different time.

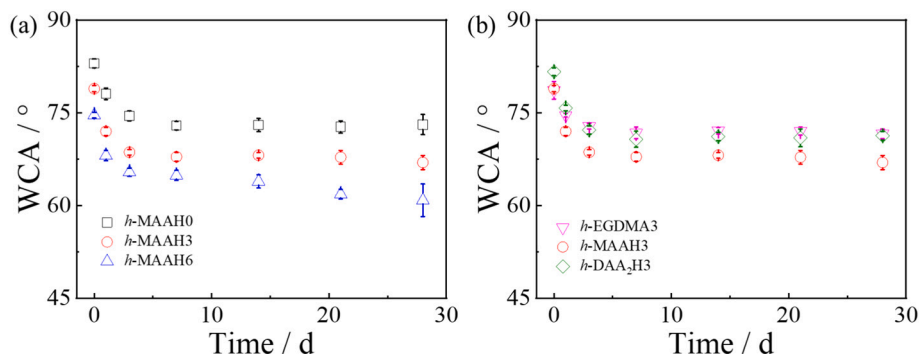


Fig. 2. Time dependence of water contact angle (WCA) of (a) *h*-MAAHx and (b) *h*-Y3 in artificial seawater.

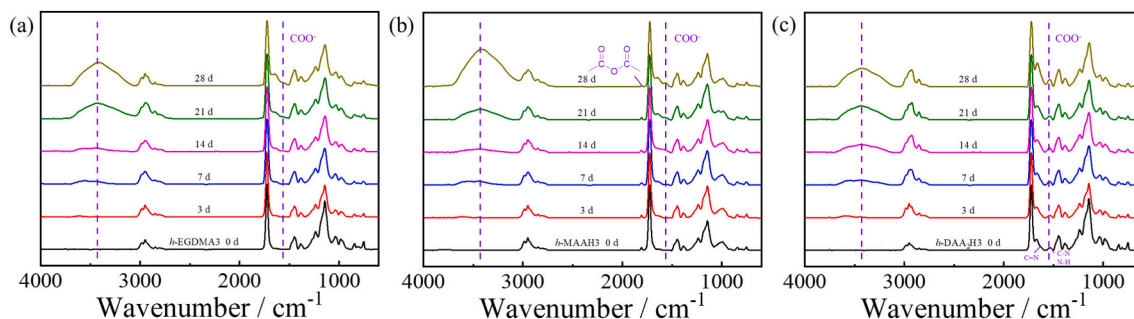


Fig. 3. ATR-FTIR spectra of (a) *h*-EGDMA3, (b) *h*-MAAH3, and (c) *h*-DAA₂H3 after immersion in artificial seawater for different time.

similar TCB contents, the peaks at 1640 and 1560 cm⁻¹ in *h*-MAAH0 are not obvious compared to *h*-MAAH3 and *h*-MAAH6 after each hydrolysis. This is probably because the MAAH moieties in the polymer can promote more hydrolysis, which is consistent with the above-mentioned change rule of WCA. For *h*-MAAH3 and *h*-MAAH6, the peak at 1805 cm⁻¹ becomes smaller during immersion in artificial seawater from 0 to 28 days, which further indicates the cleavage of the anhydride groups. Namely, the hydrolysis of the TCB groups and cleavage of the anhydride groups

can be observed during hydrolysis, respectively. In comparison, the peaks at 1640 and 1560 cm⁻¹ in *h*-EGDMA3 are not obvious compared with *h*-MAAH3 and *h*-DAA₂H3. This obscurity in the peak is because EGDMA moieties in the polymer are uncleavable and cannot contribute to additional hydrolysis groups. Compared with other samples, *h*-DAA₂H3 exhibits additional peaks at 1670, 1620, and 1530 cm⁻¹, which are attributed to the C=O stretching, C=N stretching, and C=N stretching and N-H bending of the acyl hydrazone bonds in DAA₂H [47]. The

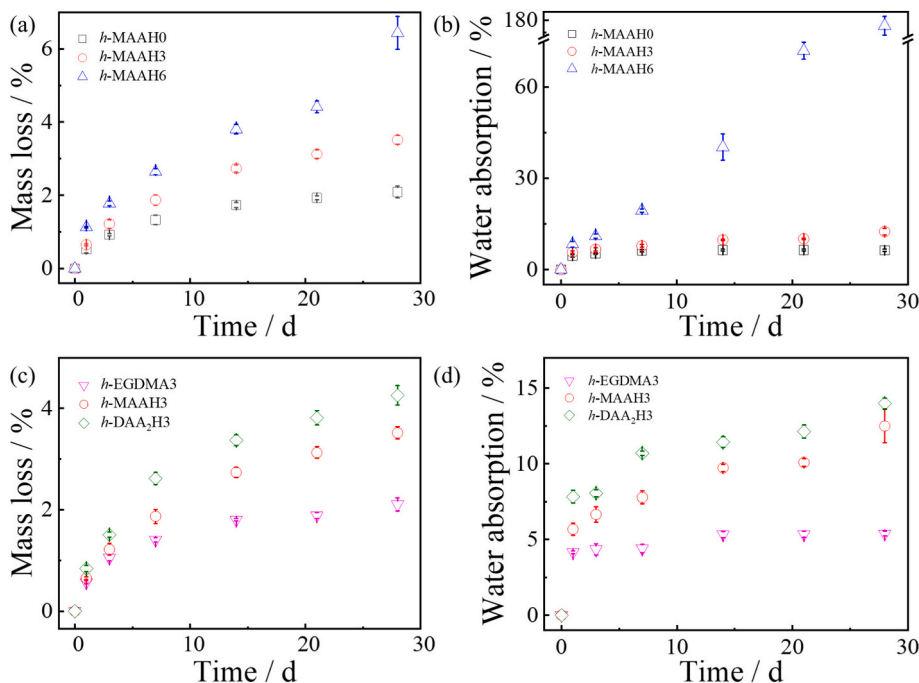


Fig. 4. Time dependence of (a, c) mass loss and (b, d) water adsorption of *h*-Yx in artificial seawater.

result indicates that the polymers contain DAA₂H fragments formed by Schiff's base reaction. According to previous reports [37,48], DAA₂H is a degradable divinyl monomer because it contains cleavable hydrazone bonds. After hydrolysis, more obvious peaks can be observed at 1640 and 1560 cm⁻¹, which are due to the synergistic effect of hydrazone bond cleavage of DAA₂H and hydrolysis of TCB groups. In short, all coating surfaces generated hydrophilic zwitterions, and the introduction of degradable branching points promotes the hydrolysis of the coatings.

We examined the degradation of the polymer coatings in artificial seawater. Fig. 4a shows the time dependence of mass loss of *h*-MAAHx after immersion in artificial seawater. The mass loss is a function of the dry weight of polymer films so that its decrease does not contain the contribution of water. All samples began to lose weight after immersion in artificial seawater for 3 days, and the mass loss increases with the immersion time. For *h*-MAAH0, it has the lowest mass loss compared to other samples. This is because its main chain has no cleavable points, and the mass loss depends only on the dissolution of the hydrophilic zwitterionic fragments generated by the hydrolysis of the TCB side groups. For *h*-MAAH3 and *h*-MAAH6, the mass loss after 28 days is ~3.5 and 6.5 wt%, respectively, which is attributed to the hydrolysis of the TCB side groups and the degradation of the cleavable branching points. In other words, once immersed in artificial seawater, the HBPs in the surface layer of coating hydrolyze and degrade into short segments or small molecule that are easily dissolved or dispersed in artificial seawater. Then, the new layer of HBPs without contact with artificial seawater is exposed and starts to hydrolyze and degrade. As immersion time goes, it keeps cycling in “exposed - hydrolyze/degrade - dissolve/disperse” mode, forming a self-generating and dynamic surface. Obviously, the degradation rate increases with the content of anhydride groups, indicating that the degradability of *h*-MAAHx can be tuned by controlling the content of the cleavage divinyl monomers. It is worth noting that the degree of branching (DB) of HBPs in the surface layer of coating after immersion becomes zero and they do not have *T_g* due to the degradation, while the underlayer of HBPs have no contact with artificial seawater so that their DB and *T_g* are still the same as the initial

values.

Fig. 4b shows the time dependence of water adsorption of *h*-MAAHx after immersion in artificial seawater. For *h*-MAAH0, the water adsorption increases from 0 to 6.3 wt% and gradually levels off with the immersion time. For *h*-MAAH3 and *h*-MAAH6, the water adsorption increases with the immersion time. The water absorption of *h*-MAAH6 is much greater than that of other samples. This is understandable because the hydrolysis of higher MAAH content generates more hydrophilic carboxyl groups, leading to more hydrophilic zwitterionic groups and higher hydration. Namely, the water adsorption of *h*-MAAHx is highly positively correlated with the MAAH content.

As shown in Fig. 4c, the mass loss of the sample *h*-EGDMA3 without cleavable branching points is similar to *h*-MAAH0. For *h*-MAAH3 and *h*-DAA₂H3, the mass loss values are ~3.5 and 4.3 wt% after 28 days, respectively. This fact indicates that the degradation rate of *h*-DAA₂H3 is greater than that of *h*-MAAH3. This may be because a DAA₂H monomer contains two cleavable hydrazone bonds, which is more conducive to artificial seawater attacking the cleavage points and promoting polymer degradation. In addition, the hydrolysis of DAA₂H also produces small molecule amines that are easily soluble in water. After degradation, more water-soluble fragments and amines will be generated, and the hydrolysis of the TCB groups will generate zwitterions. The synergistic effect of the two promotes the self-renewal of the coating surface. It is worth noting that the roughness of *h*-DAA₂H3 (Fig. 5b) is much larger than other samples, which increases the diffusion of water and facilitates the attack to hydrazone bond. As shown in Fig. 4d, the water absorption of all samples gradually increases with the immersion time, which is consistent with the change in mass loss. In short, we can adjust the degradation rate of coatings by tuning the content or type of divinyl monomers. The degradability of the coatings from large to small is *h*-MAAH6 > *h*-MAAH3 > *h*-MAAH0 and *h*-DAA₂H3 > *h*-MAAH3 > *h*-EGDMA3.

We also conducted the surface morphology tests to evaluate the surface properties of the coatings. Fig. 5 shows the roughness of the polymers and the typical surface morphology of *h*-MAAH3 with different

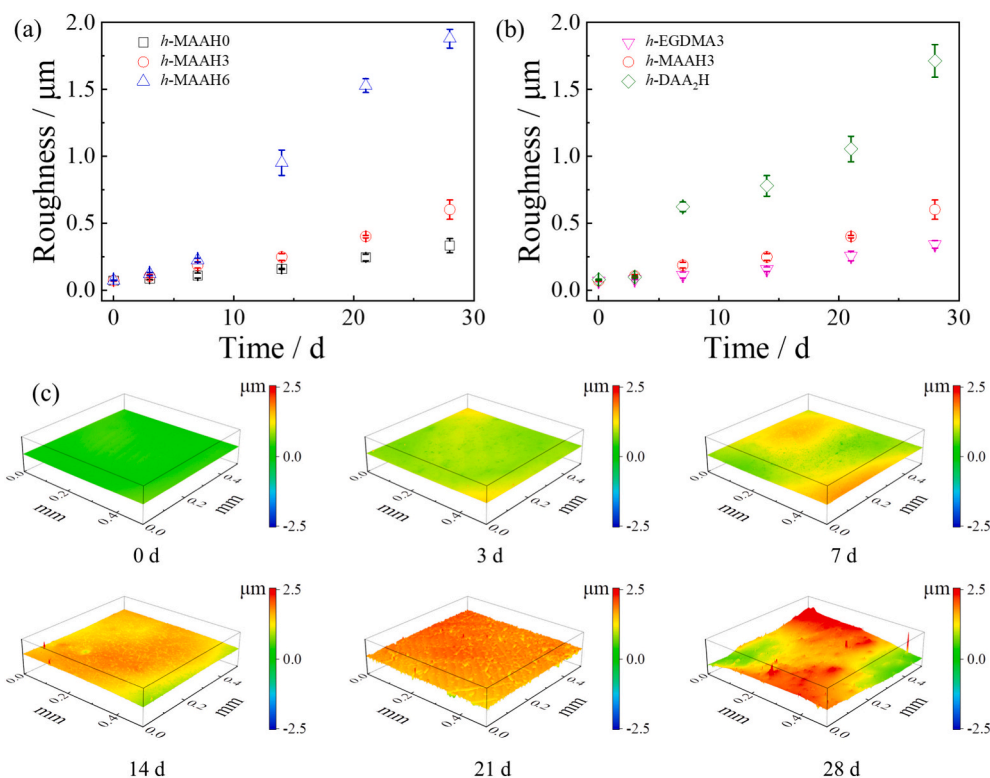


Fig. 5. Surface roughness of (a) *h*-MAAHx and (b) *h*-Y3 with different immersion time in artificial seawater; and (c) typical surface morphology of *h*-MAAH3.

immersion time. The surface roughness of all samples is nanometer level, and their surface is smooth. The roughness of *h*-DAA₂H3 is ~ 80 nm, and the roughness of other samples is ~ 70 nm, indicating that the type or content of divinyl monomers have little effect on the surface roughness. The roughness of all samples increases with the immersion time, indicating that the hydrolysis of the coating surface would result in the morphology change. For *h*-MAAH0 and *h*-EGDMA3, the surface roughness changes from ~ 0.07 to $0.33 \mu\text{m}$ after immersion for 28 days. This is understandable because they do not contain cleavable branching points, and degradation only relies on TCB groups. The surface roughness of *h*-MAAH3 gradually increases from ~ 0.07 to $0.57 \mu\text{m}$ after 28 days. However, after 14 days of immersion in artificial seawater, the surface roughness of *h*-MAAH6 is much greater than that of *h*-MAAH3 and gradually reaches $\sim 1.88 \mu\text{m}$ after 28 days. This increase in surface roughness of *h*-MAAH6 indicates that its degradability is better than that of *h*-MAAH3, which is consistent with the mass loss results (Fig. 4). Using *h*-MAAH3 (Fig. 5c) as a representative, it exhibits the surface morphology changes from smooth to rough during hydrolysis. This is because the faster degradation of anhydride groups results in a rougher surface. For *h*-DAA₂H3, the appearance of a rough surface can also be observed, and its roughness is as high as $\sim 1.71 \mu\text{m}$ after 28 days of immersion in artificial seawater, which is caused by the rapid degradation of the hydrazone bonds in artificial seawater. Note that even if the roughness continues to increase, all surfaces reach a constant WCA after 14 days (Fig. 2), this is because the composition and structure of the polymers have a greater impact on the WCA than the surface morphology in this system.

Coatings or surfaces that can inhibit bacterial adhesion in the long-term are essential for anti-biofouling applications. Herein, we used marine bacteria as a model to investigate the long-term anti-biofouling performance of polymer surfaces. Fig. 6 shows the fluorescent images of *Pseudomonas* sp. and the relative bacteria adhesion (RBA) on different surfaces. At any time, a large crowd of bacteria (RBA $\sim 100\%$) can be observed on silicon wafers, indicating that the fouling pressure is high. For all samples without hydrolysis, a large number of bacteria (RBA $> 90\%$) can be observed on the surface because of the lack of anti-biofouling moieties on the surfaces. After immersion in artificial seawater for 3 days, there are still many bacteria attached to the

surfaces. For *h*-MAAH0, *h*-MAAH3, and *h*-EGDMA3, the RBA is $\sim 80\%$, indicating that the surface cannot resist bacterial adhesion. This is because the hydrolysis of the TCB group is slow and the surface does not generate enough zwitterions. For *h*-MAAH6 and *h*-DAA₂H3, the RBA is $\sim 63\%$. Compared with other samples, the two coatings are slightly resistant to the adhesion of bacteria. This is because the separate degradation of anhydride groups and hydrazone bonds can promote the hydrolysis of TCB, resulting in a slight increase in zwitterions. After immersion in artificial seawater for 7, 14, 21, and 28 days, almost no bacteria were observed in all samples, respectively. These results indicate that the hydrolysis of TCB groups can generate enough zwitterions to resist bacterial adhesion. In summary, all coatings exhibit excellent and stable anti-biofouling performance during hydrolysis due to the similar contents of TCB.

Fig. 7 shows the optical images of the diatom *Navicula incerta* on polymer surfaces with their density after immersion in artificial seawater for different time. A great number of diatoms ($\sim 340 \text{ cell}\cdot\text{mm}^{-2}$) can always be observed on silicon wafers with different immersion time. For all samples, the density of diatoms before immersion in artificial seawater is greater than $320 \text{ cell}\cdot\text{mm}^{-2}$, indicating that surfaces are easily colonized by diatoms. After immersion in artificial seawater for 3 days, the densities of diatoms are $\sim 225, 240, 212, 234$, and $234 \text{ cell}\cdot\text{mm}^{-2}$ on *h*-MAAH0, 3, 6, and *h*-EGDMA3, *h*-DAA₂H3 surfaces, respectively. After 7 days, the density of diatoms in all samples decreased. Diatoms tend to colonize on hydrophobic surfaces [7]. Thus, the decreased density of diatoms can be attributed to the hydrolysis of the TCB groups, which generates zwitterions and changes surface hydrophilicity. The diatoms are barely visible on the surface after immersion in artificial seawater for 7, 14, 21, and 28 days. The similar anti-diatom performance for all samples further indicates that the stability of fouling resistance. Similar to the above antibacterial result, all the polymers exhibit excellent anti-diatom performance. In short, these facts further indicate that the hydrolysis of TCB can prevent biofouling effectively.

5. Conclusions

We have prepared self-regenerating zwitterionic hyperbranched

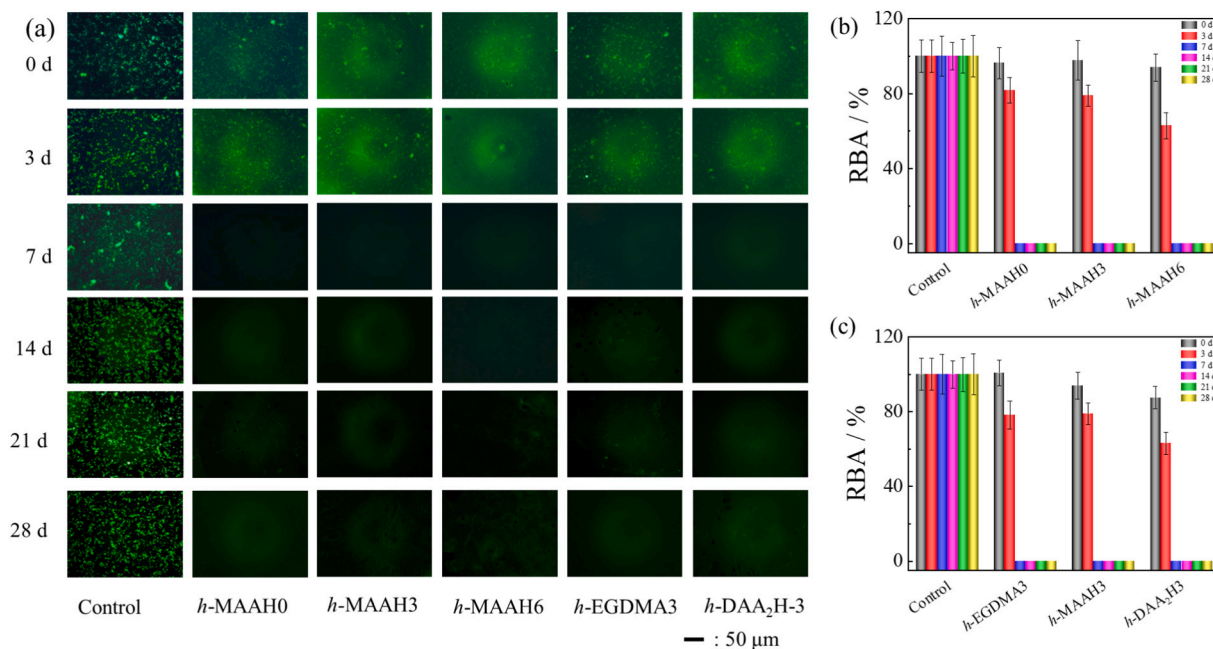


Fig. 6. (a) Fluorescence photographs of *Pseudomonas* sp. adhered on different surfaces with different immersion time in artificial seawater, and the relative bacterial adsorption (RBA) of (b) *h*-MAAHx and (c) *h*-Y3.

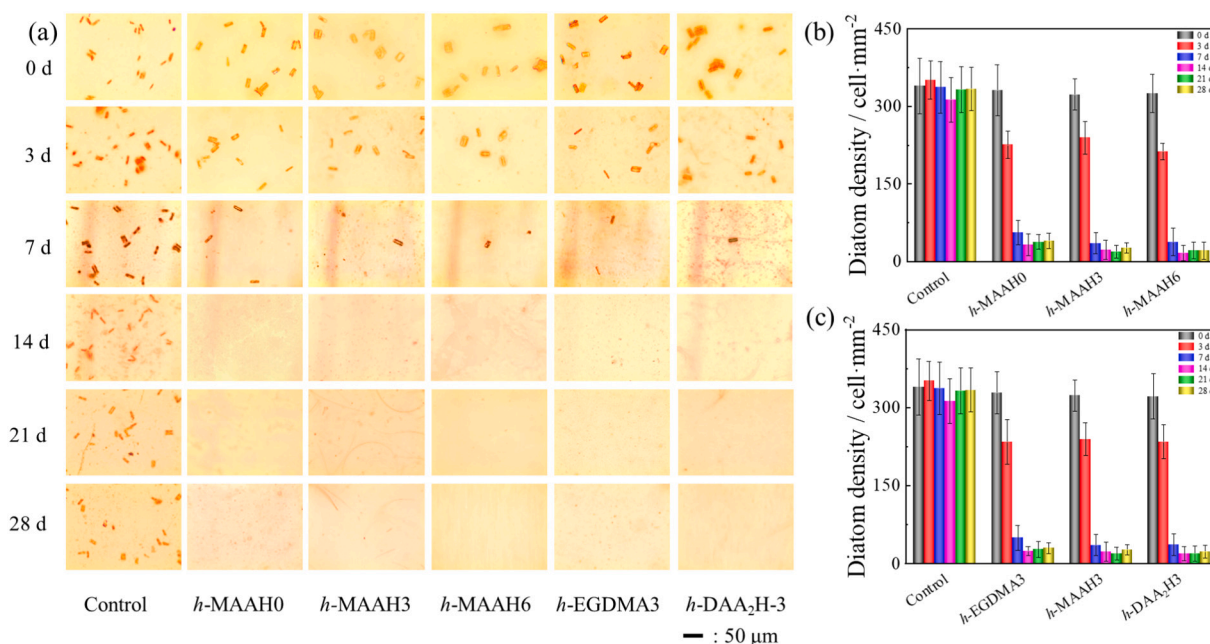


Fig. 7. (a) Optical microscopy images of the diatom *Navicula incerta* on polymer surfaces with different immersion time in artificial seawater, and the density of diatoms on polymer surfaces of (b) *h*-MAAH x and (c) *h*-Y3.

polymers via RAFT polymerization of methyl methacrylate, tertiary carboxybetaine ester (TCB), and divinyl monomers. The hydrolytic degradation shows that the degradation rate of such polymers is controlled by the content or type of divinyl monomers in primary chains. Moreover, the synergy of branching points degradation and TCB side groups hydrolysis enables the polymers to be degraded into small fragments in artificial seawater, creating a self-renewal surface with anti-biofouling function, exhibiting excellent antibacterial and anti-diatom properties. Such polymeric coatings can be used in marine anti-biofouling.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2021.106674>.

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